

Milestone 3: Panorama Generation Prototype (M3)

Project goal: deliver an end-to-end multi-image panorama stitching pipeline that upgrades a 2-image ORB baseline into a robust N-image prototype capable of reliable, high-accuracy feature matching and seamless compositing.

Dataset (M3): self-collected handheld phone images 12MP, ~30–50% horizontal overlap, variable lighting between consecutive frames. Objective: demonstrate scalable N-image stitching with improved geometric accuracy and acceptable runtime.

[GitHub Repository :](#)



Group members:

E/22/054

E/22/184

E/22/179

E/22/205



Working Method: Feature Extraction & Matching

SIFT for Descriptors

Replaced ORB with SIFT:
128-d floating-point descriptors that are robust to scale, rotation, and illumination shifts critical for handheld captures with varying exposures.



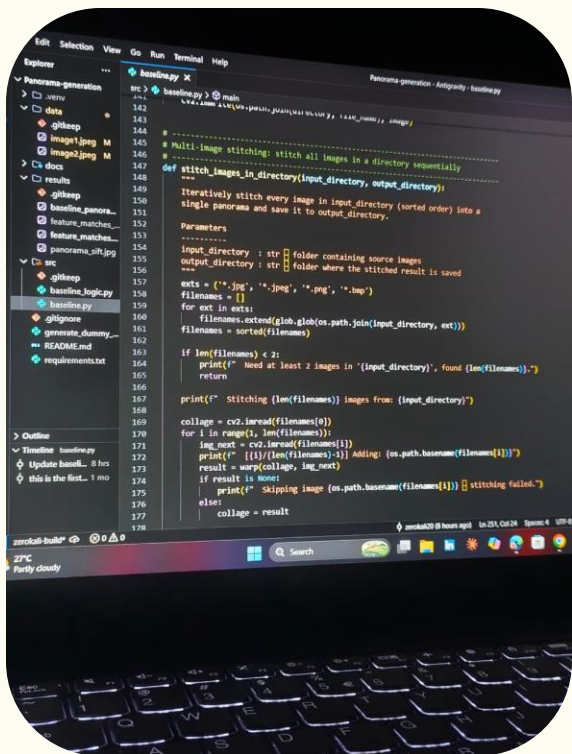
FLANN Matcher (KD-Trees)

FLANN maintains high-speed matching at SIFT's dimensionality via approximate KD-tree search balances throughput with match quality for N-image pipelines.

Filtering: Lowe's Ratio & Quality Cutoffs

Applied Lowe's ratio test (threshold 0.7) to aggressively reject ambiguous background matches and reduce downstream RANSAC failures.

Implementation: Iterative Stitching Pipeline



Architecture summary, an automated sequential loop that ingests a sorted directory of N images and stitches them into a growing canvas.

- **Start:** Image 1 initializes the base canvas.
- **Per incoming image:**
 - compute feature matches → estimate Homography via RANSAC → deduce warped image bounding box.
- **Canvas expansion:** allocate new canvas region to fit warped image, then warp and composite into canvas space.
- Repeat iteratively for N images, preserving ordering to limit inter-frame parallax.



Key engineering choices:
sorted image order, aggressive outlier rejection,
dynamic canvas allocation to avoid cropping.

Quantitative Evaluation Metrics

Execution Speed

~0.65 seconds per image pair (SIFT + FLANN). Performance optimized for 600px-height resized evaluation images.

Feature Robustness (Inlier Ratio)

384 raw matches → 294 inliers after RANSAC
(76.56%) substantial improvement over the M2 ORB baseline.

Geometric Accuracy (RMSE)

Root Mean Square Reprojection Error (inliers):
0.54 pixels sub-pixel alignment that prevents structural ghosting.

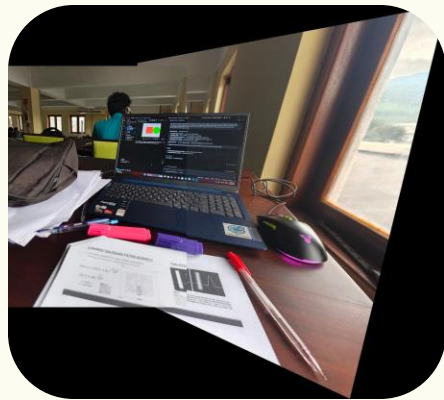
```
Done.  
● PS D:\Academic\Semester 04> python -u "  
te_friend.py"  
Total Matches: 384  
Inliers: 294 (76.56%)  
RMSE: 0.5468 pixels  
Time: 0.7507 seconds  
❖ PS D:\Academic\Semester 04>
```

Qualitative Results: Stitched Panorama

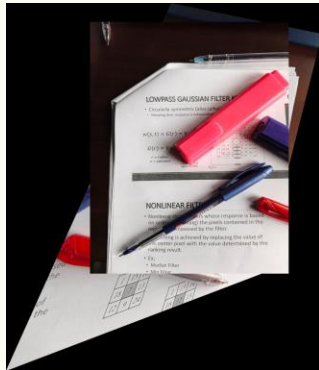
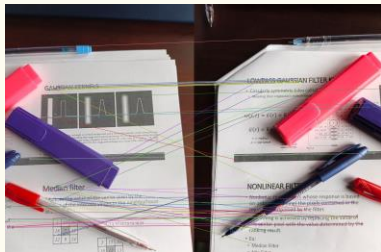
Representative result:

multi-image panorama demonstrating preserved structural lines and correct horizon alignment. Sub-pixel homography accuracy prevents visible ghosting; dynamic bounding-box logic avoids cropping of content near edges.

- 📄 Visual inspection confirms alignment quality but reveals visible exposure seams in overlaps.



Failure Cases & Observations



- **Visible seam lines:** Current "hard overwrite" compositing shows abrupt exposure transitions where frames differ in brightness.
- **Unconstrained canvas / catastrophic warp:** When RANSAC accepts a poor homography (extreme perspective skew), the warped image stretches and can cause memory blowouts or visually distorted regions.
- **Iterative drift:** Accumulation of small sub-pixel errors across 4+ sequential stitches leads to noticeable misalignment over longer panoramas.



- ⚠ These failure modes are reproducible on handheld sequences with strong perspective changes or inconsistent exposure.

Planned Improvements (Milestone 4)

Advanced Blending

Implement multi-band pyramidal blending or distance-based feathering (alpha gradients) to remove visible seams while preserving high-frequency detail.

Smart Fallback Transform

Add a safety check on perspective skew: if stretching $> 1.3\times$ width, switch from Homography to a constrained Affine fallback to prevent catastrophic warps.

Global Bundle Adjustment

Perform global optimization (bundle adjustment) over all camera poses to mitigate iterative drift and enforce global geometric consistency across the panorama.

Additional Engineering Tasks

Integrate exposure compensation, seam-finding masks, and confidence-based compositing. Add unit tests and a small benchmark suite on varied handheld captures.

- ❏ close failure modes while maintaining runtime targets and validating improvements on the same self-collected dataset.

